Cash Flow Analysis

When Payment Period Is Equal to Compounding Period

Whenever the compounding and payment periods are equal whether the interest is compounded annually or at some other interval, the following solution method can be used:

1. Identify the number of compounding periods (M) per year.

2. Compute the effective interest rate per payment period. Then, with C = 1 and K = M we have,

$$i = \frac{r}{M}.$$

3. Determine the total number of compounding periods:

$$N = M \times (\text{number of years}).$$

Suppose you make an annual contribution of \$3,000 to your savings account at the end of each year for 10 years. If the account earns 7% interest annually, how much can be withdrawn at the end of 10 years (Figure 3.19)?

n = No. of years, M = Compounding Frequency per year = 1, K = Payment Frequency per year = 1,

N = Total payment periods = K *n = 10 times

A = \$3000, m=1, n= 10, K = 1, N= 10, r = 7%/year, F = ?

$i_c = r/m = 7\%/1 = 7\%/yr.$

$$F = A\left[\frac{(1+i)^N - 1}{i}\right] = A(F/A, i, N).$$

$i_e = [1+ r/m]^m - 1$

$i_e = [1+ 0.07] - 1$

$i_e = 0.07/yr$ OR 7%/yr.

$i = [1+ r/m]^c - 1 = 0.07/yr$ OR 7%/yr.

Suppose you make an annual contribution of \$3,000 to your savings account at the end of each year for 10 years. If the account earns 7% interest annually, how much can be withdrawn at the end of 10 years (Figure 3.19)?

Given: $A = \$3{,}000$, $N = 10$ years, and $i = 7\%$ per year.
Find: F.

M = Compounding Frequency = 1, K = Payment Frequency per year = 1,
N = Total payment periods = K * No. of Years = 10 times

$$F = A\left[\frac{(1+i)^N - 1}{i}\right] = A(F/A, i, N).$$

F = 3000 (F/A, 0.07, 10)

F = 3000 (13.8164)

F = 41, 449.

Compounding Is More Frequent than Payments

The computational procedure for compounding periods and payment periods that cannot be compared is as follows:

1. Identify the number of compounding periods per year (*M*), the number of payment periods per year (*K*), and the number of interest periods per payment period (*C*).

2. Compute the effective interest rate per payment period: For discrete compounding, compute:

$$i = (1+r/m)^c - 1$$

3. Find the total number of payment periods:

$$N = K \times (\text{number of years}).$$

4. Use *i* and *N* in the appropriate formulas.

Compounding Is More Frequent than Payments

Suppose you make equal quarterly deposits of $1,500 into a fund that pays interest at a rate of 6% compounded monthly. Find the balance at the end of year 2.

$$F = A\left[\frac{(1+i)^N - 1}{i}\right] = A(F/A, i, N).$$

Compounding Is More Frequent than Payments

Suppose you make equal quarterly deposits of \$1,500 into a fund that pays interest at a rate of 6% compounded monthly. Find the balance at the end of year 2.

Given: A = \$1,500 per quarter, r = 6% per year, M = 12 compounding periods per year, and N = 8 quarters.

$$F = A\left[\frac{(1+i)^N - 1}{i}\right] = A(F/A, i, N).$$

Suppose you make equal quarterly deposits of $1,500 into a fund that pays interest at a rate of 6% compounded monthly, Find the balance at the end of year 2.

We follow the aforementioned procedure for noncomparable compounding and payment periods:

1. Identify the parameter values for M, K, and C, where

$$
\begin{aligned}
M &= 12 \text{ compounding periods per year,} \\
K &= 4 \text{ payment periods per year,} \\
C &= 3 \text{ interest periods per payment period.}
\end{aligned}
$$

c = m/K

2. Use Eq. (4.2) to compute the effective interest:

$$
\begin{aligned}
i &= \left(1 + \frac{0.06}{12}\right)^3 - 1 \\
&= 1.5075\% \text{ per quarter.}
\end{aligned}
$$

i = (1 + r/m)^c

3. Find the total number of payment periods, N:

$$N = K(\text{number of years}) = 4(2) = 8 \text{ quarters.}$$

4. Use i and N in the appropriate equivalence formulas:

$$F = \$1{,}500(F/A, 1.5075\%, 8) = \$12{,}652.60$$

Practice Question

A series of equal quarterly payments of $5,000 for 12 years is equivalent to what future amount at an interest rate of 9% compounded monthly?

$$F = A\left[\frac{(1+i)^N - 1}{i}\right] = A(F/A, i, N).$$

Uniform Gradient Cash flow Series

- An **arithmetic gradient** series is a cash flow series that either increases or decreases by a **constant amount** each period. The amount of change is called the **gradient**.
- Assume a manufacturing engineer predicts that the cost of maintaining a robot will increase by $5000 per year until the machine is retired. The cash flow series of maintenance costs involves a constant gradient, which is $5000 per year.

Uniform Gradient Cash flow Series

- For example, if you purchase a used car with a 1-year warranty, you might expect to pay the gasoline and insurance costs during the first year of operation. Assume these cost $2500; that is, $2500 is the base amount. After the first year, you absorb the cost of repairs, which can be expected to increase each year. If you estimate that total costs will increase by $200 each year, the amount the second year is $2700, the third $2900, and so on to year n , when the total cost is 2500 (n 1)200. Note that the gradient ($200) is fi rst observed between year 1 and year 2, and the base amount ($2500 in year 1) is not equal to the gradient.
- Define the symbols G for gradient and CF_n for cash flow in year n as follows.

G = constant arithmetic change in cash flows from one time period to the next; G may be positive or negative.

$$CF_n = \text{base amount} + (n - 1)G \qquad [2.18]$$

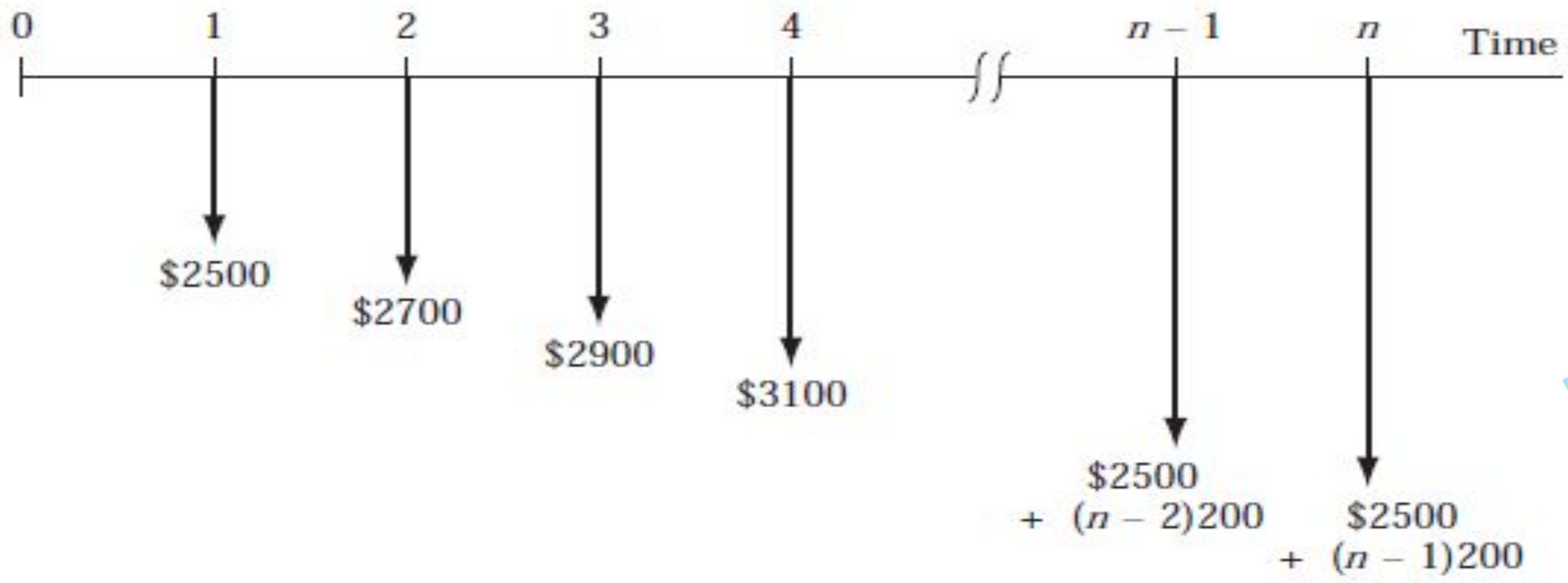


Figure 2–11
Cash flow diagram of an arithmetic gradient series

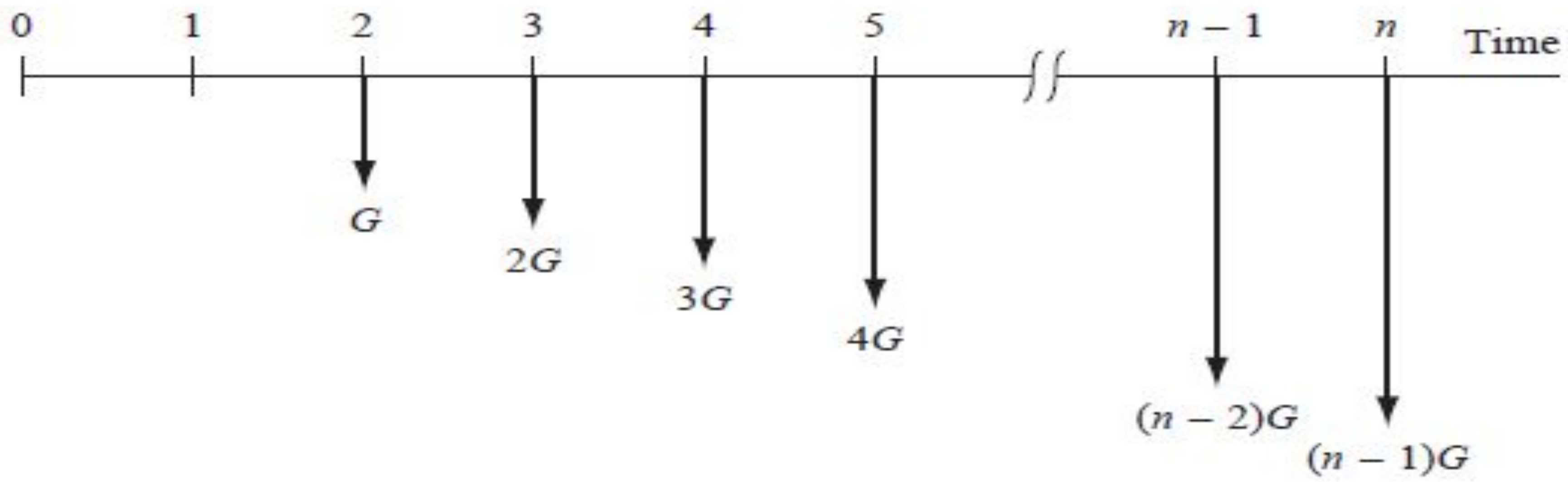


Figure 2–12
Conventional arithmetic gradient series without the base amount.

Estimated revenues of a local company are $80,000 for first year. Expected to rise by uniform amount to a total of $200,000 by the end of year 9.

a. Determine the Gradient.
b. What will be the firm's revenue in year 5.

Solution: All values are taken in $1000 unit.

a. G = Slope = Δ Cash flow/ Δ years = (200 – 80)/ (9-1) = 120/8 = 15 OR $15000

b. $CF_n = A_0 + (n-1)G$

$CF_5 = 80 + (5-1)\ (15)$

$CF_5 = 80 + (4)(15) = 140$ OR $140, 000

Notice that $80,000 is the revenue earned in year 1, $140, 000 is the revenue earned in only year 5, similarly $200,000 is the revenue earned in year 9. Total cash flow during the 9-year period will be the sum of all annual revenue flows occurred during the given period

Arithmetic gradient to annuity conversion factor

The **arithmetic gradient to annuity conversion factor**, denoted by ($A/G,i,N$), gives the value of an annuity, A, that is equivalent to an arithmetic gradient series where the constant increase in receipts or disbursements is G per period, the interest rate is i *defined for per payment period*, and the *total number of payment periods* is N.

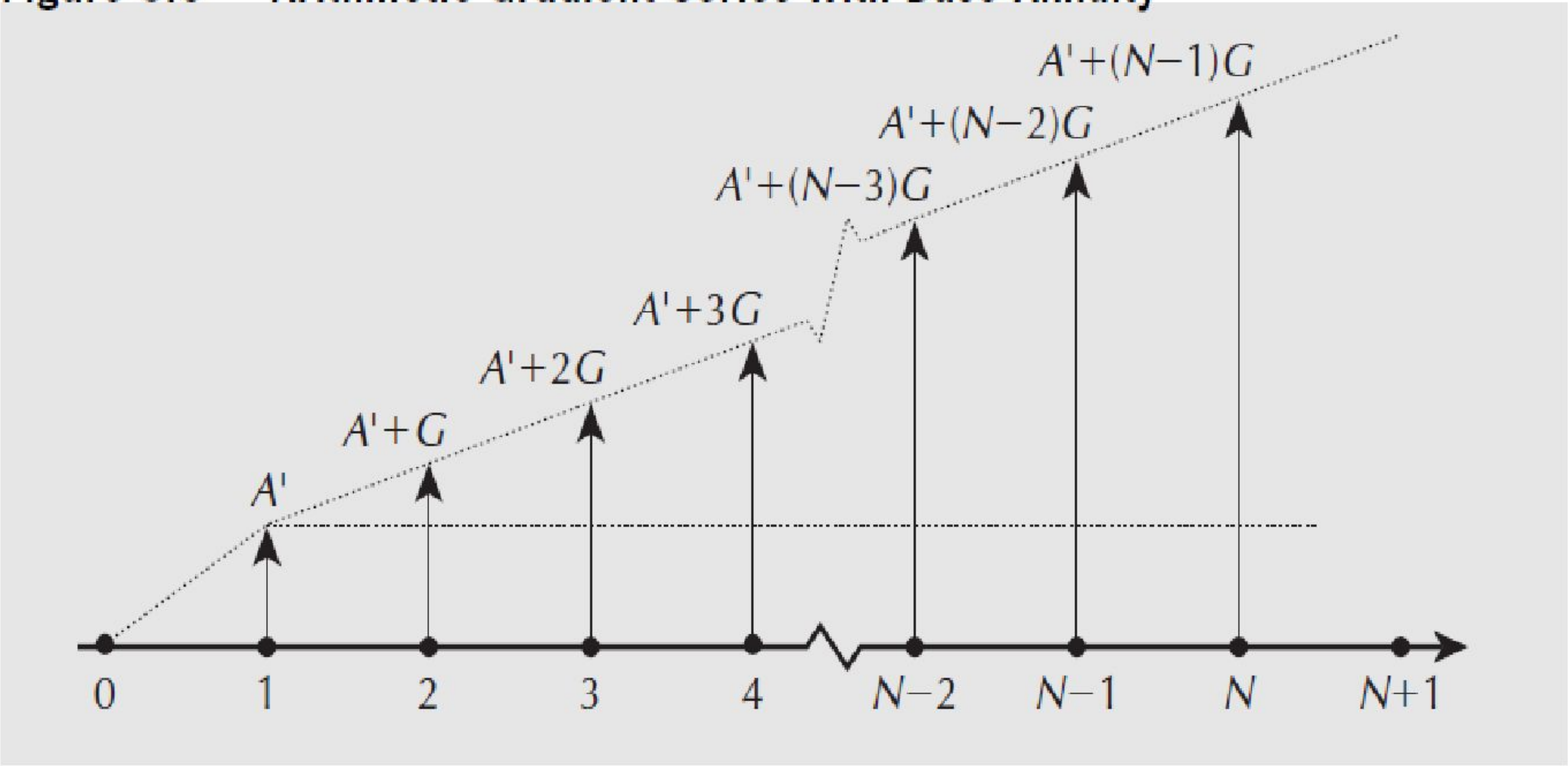
$$(A/G,i,N) = \frac{1}{i} - \frac{N}{(1+i)^N - 1}$$

Arithmetic gradient to annuity conversion factor

There is often a base annuity A_o associated with a gradient, as illustrated in Figure 3.6. To determine the uniform series equivalent to the *total* cash flow, the base annuity A_o must be included to give the overall annuity:

$$A = A_{Tot} = A_0 + G(A/G, i, N)$$

Figure 3.6 Arithmetic Gradient Series With Base Annuity



A person is planning for his retired life. He has 10 more years of service. He would like to deposit 20% of his salary, which is Rs. 4,000, at the end of the first year, and thereafter he wishes to deposit the amount with an annual increase of Rs. 500 for the next 9 years with an interest rate of 15%. Find the total amount at the end of the 10th year of the above series.

$$(A/G,i,N) = \frac{1}{i} - \frac{N}{(1+i)^N - 1}$$

Solution Here,

$A1$ = Rs. 4,000 base amount

G = Rs. 500 the gradient on payment/deposit

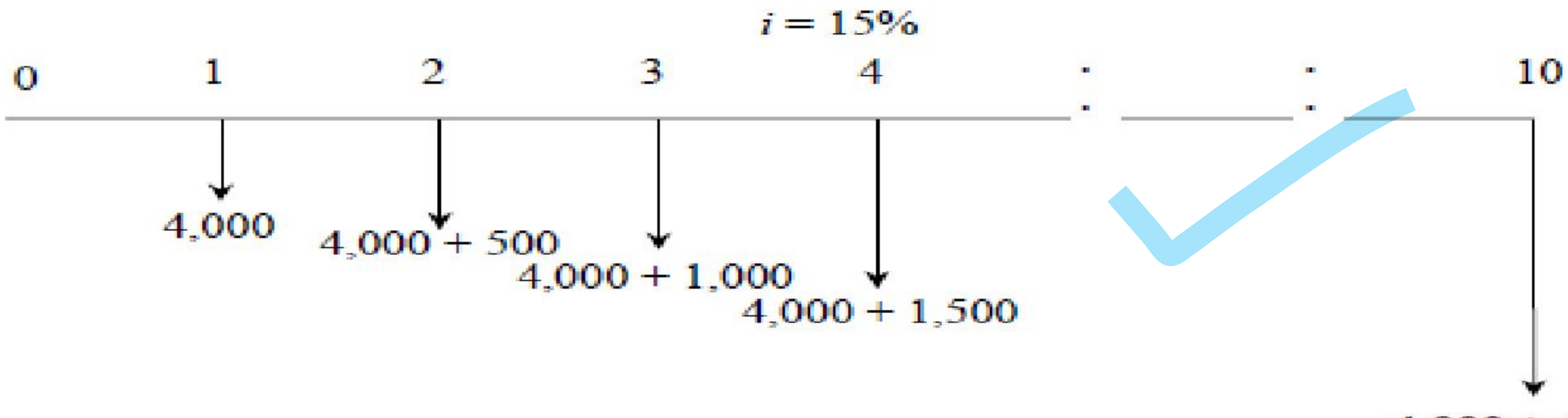
i = 15%

n = 10 years

A = ? & F = ?

$$(F/A,i,N) = \frac{(1+i)^N - 1}{i}$$

The cash flow diagram is shown in Fig. 3.13.



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$$A = A_0 + G(A/G, i, N)$$

$$= 4{,}000 + 500(A/G,\ 15\%,\ 10)$$

$$= 4{,}000 + 500 \times 3.3832$$

$$= \text{Rs. } 5{,}691.60$$

$$(F/A,i,N) = \frac{(1+i)^N - 1}{i}$$

$$F = A\ (F/A,\ i,\ N)$$

$$= A(F/A,\ 15\%,\ 10)$$

$$= 5{,}691.60(20.304)$$

$$= \text{Rs. } 1{,}15{,}562.25$$

A person is planning for his retired life. He has 10 more years of service. He would like to deposit Rs. 8,500 at the end of the first year and thereafter he wishes to deposit the amount with an annual decrease of Rs. 500 for the next 9 years with an interest rate of 15%. Find the total amount at the end of the 10th year of the above series.

$$(A/G,i,N) = \frac{1}{i} - \frac{N}{(1+i)^N - 1}$$

$$(F/A,i,N) = \frac{(1+i)^N - 1}{i}$$

A person is planning for his retired life. He has 10 more years of service. He would like to deposit Rs. 8,500 at the end of the first year and thereafter he wishes to deposit the amount with an annual decrease of Rs. 500 for the next 9 years with an interest rate of 15%. Find the total amount at the end of the 10th year of the above series.

$$A = A_0 + G(A/G, i, N)$$

$$= 8{,}500 - 500(A/G, 15\%, 10)$$

$$= 8{,}500 - 500 \times 3.3832$$

$$= \text{Rs. } 6{,}808.40$$

$$F = A\,(F/A, i, N)$$

$$= A(F/A, 15\%, 10)$$

$$= 6{,}808.40(20.304)$$

$$= \text{Rs. } 1{,}38{,}237.75$$

Practice Question

Neighboring parishes in Louisiana have agreed to pool road tax resources already designated for bridge refurbishment. At a recent meeting, the engineers estimated that a total of $500,000 will be deposited at the end of next year into an account for the repair of old and safety-questionable bridges throughout the area. Further, they estimate that the deposits will increase by $100,000 per year for only 9 years thereafter, then cease. Determine the equivalent (*a*) present worth and (*b*) annual series amounts, if public funds earn at a rate of 5% per year.

Geometric Gradient Series

A geometric gradient series is a series of cash flows that increase or decrease by a constant *percentage* each period. The geometric gradient series may be used to model inflation or deflation, productivity improvement or degradation, and growth or shrinkage of market size, as well as

In a geometric series, the base value of the series is A and the "growth" rate in the series (the rate of increase or decrease) is referred to as g. The terms in such a series are given by A, $A(1 + g)$, $A(1 + g)^2$, . . . , $A(1 + g)^{N-1}$ at the ends of periods 1, 2, 3, . . . , N, respectively. If the rate of growth, g, is positive, the terms are increasing in value. If the rate of growth, g, is negative, the terms are decreasing. Figure 3.7

Figure 3.7 Geometric Gradient Series for Receipts With Positive Growth

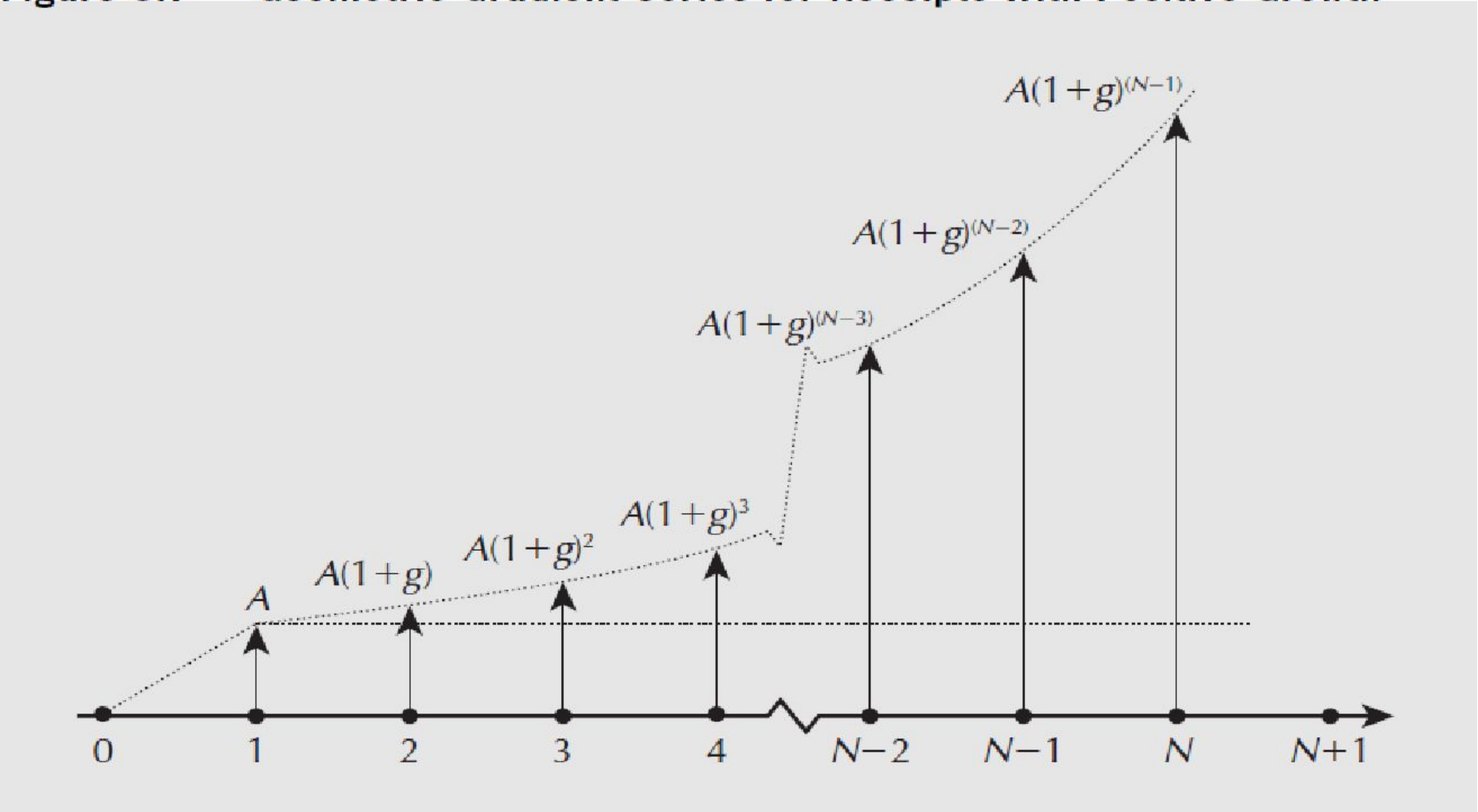
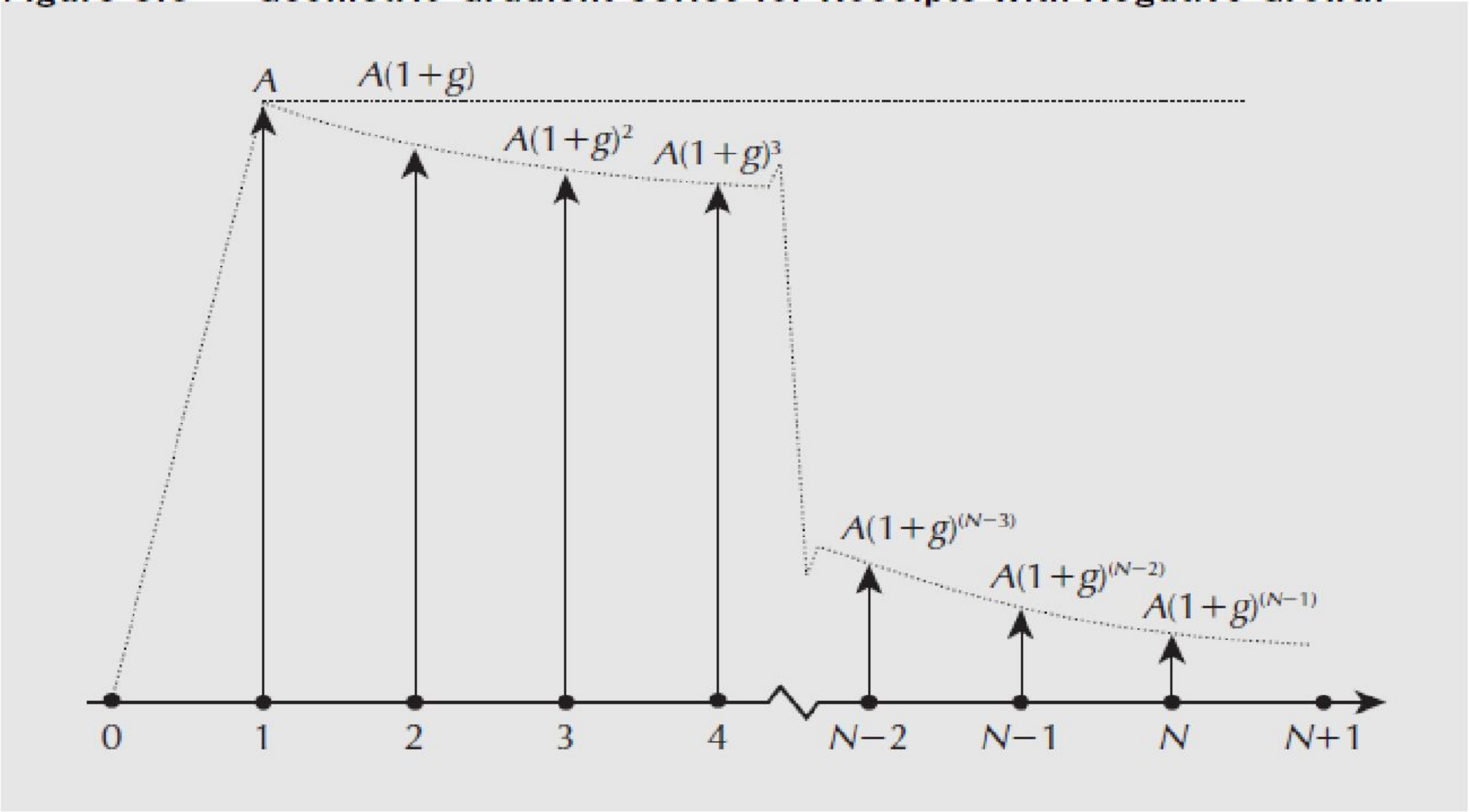


Figure 3.8 Geometric Gradient Series for Receipts With Negative Growth



Conversion Factor for Geometric Gradient Series

The **geometric gradient to present worth conversion factor**, denoted by ($P/A,g,i,N$), gives the present worth, P, that is equivalent to a geometric gradient series where the base receipt or disbursement is A, and where the rate of growth is g, the interest rate is i *per payment period*, and *total number of payment periods* is N.

The equation for P_g and the $(P/A,g,i,n)$ factor formula are

$$P_g = A_1(P/A,g,i,n)$$

$$(P/A,g,i,n) = \begin{cases} \dfrac{1 - \left(\dfrac{1+g}{1+i}\right)^n}{i-g} & g \neq i \\[2ex] \dfrac{n}{1+i} & g = i \end{cases}$$

A coal-fired power plant has upgraded an emission control valve. The modification costs only \$8000 and is expected to last 6 years with a \$200 salvage value. The maintenance cost is expected to be high at \$1700 the first year, increasing by 11% per year thereafter. Determine the equivalent present worth of the modification and maintenance cost by hand and by spreadsheet at 8% per year.

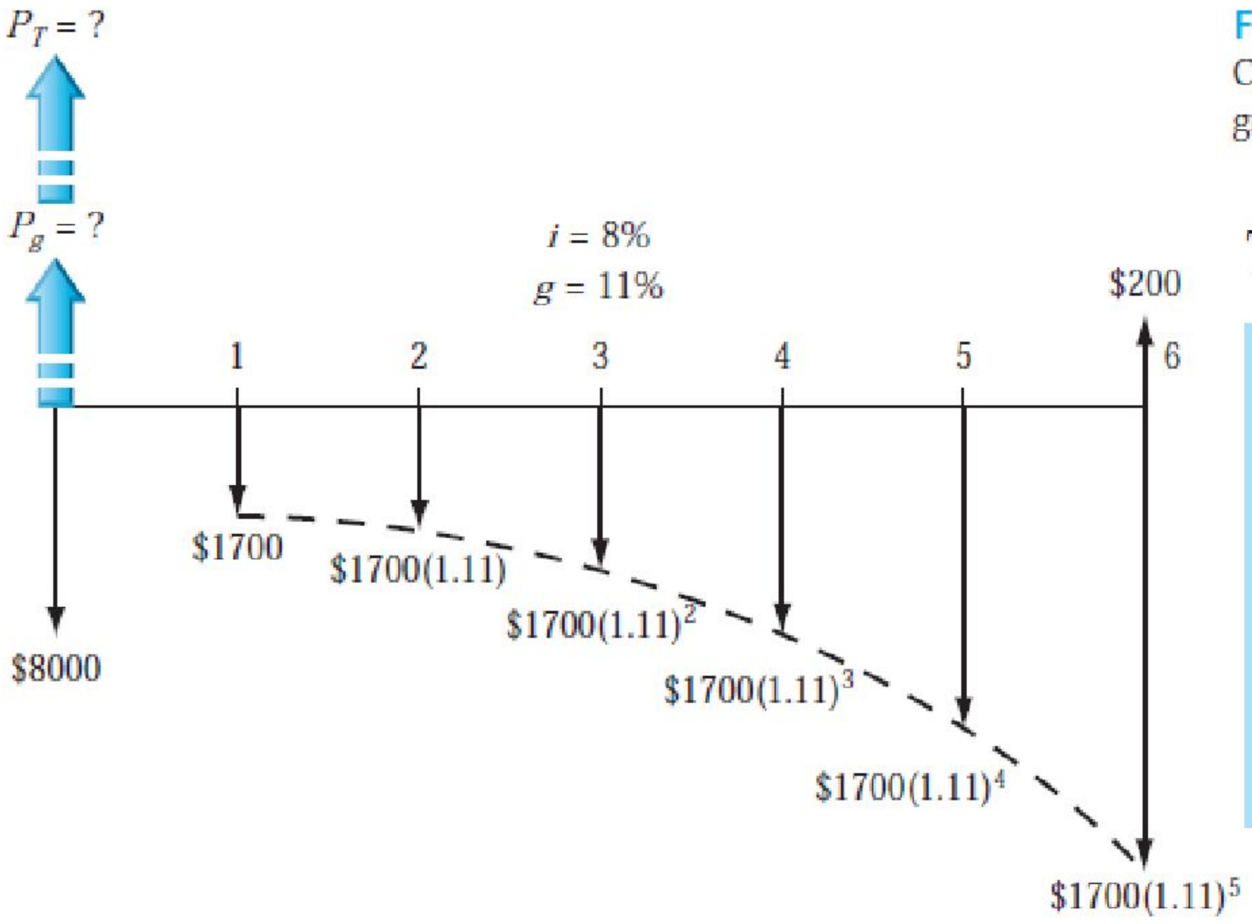


Figure 2–22
Cash flow diagram of a geometric gradient, Example 2.11.

The equation for P_g and the $(P/A,g,i,n)$ factor formula are

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The cash flow shows the salvage value as a positive cash flow and all costs as negative. Total P_T is the sum of three present worth components. For $g \neq i$ to calculate P_g.

$$P_T = -8000 - P_g + 200(P/F,8\%,6)$$

$$= -8000 - 1700\left[\frac{1-(1.11/1.08)^6}{0.08-0.11}\right] + 200(P/F,8\%,6)$$

$$= -8000 - 1700(5.9559) + 126 = \$-17{,}999$$

The equation for P_g and the $(P/A,g,i,n)$ factor formula are

$$P_g = A_1(P/A,g,i,n)$$

$$(P/A,g,i,n) = \begin{cases} \dfrac{1-\left(\dfrac{1+g}{1+i}\right)^n}{i-g} & g \neq i \\ \dfrac{n}{1+i} & g = i \end{cases}$$

Assume that revenue may start at \$50 million by the end of the first year, but then decreases geometrically by 12% per year through year 5. Determine the **present worth** and **future worth** equivalents of all revenues during this 5-year time frame at the same rate used previously, that is, 10% per year.

$$P_g = A_1(P/A,g,i,n)$$

$$(P/A,g,i,n) = \begin{cases} \dfrac{1 - \left(\dfrac{1+g}{1+i}\right)^n}{i-g} & g \neq i \end{cases}$$

Assume that revenue may start at \$50 million by the end of the first year, but then decreases geometrically by 12% per year through year 5. Determine the **present worth** and **future worth** equivalents of all revenues during this 5-year time frame at the same rate used previously, that is, 10% per year.

First, we determine P_g in year 0 using with i=0.10 and g=0.12, then we calculate F in year 5. In \$1 million units,

$$P_g = A_1(P/A,g,i,n)$$

$$(P/A,g,i,n) = \begin{cases} \dfrac{1-\left(\dfrac{1+g}{1+i}\right)^n}{i-g} & g \neq i \end{cases}$$

$$P_g = 50\left[\frac{1-\left(\frac{0.88}{1.10}\right)^5}{0.10-(-0.12)}\right] = 50[3.0560]$$

$$= \$152.80$$

$$F = 152.80(F/P,10\%,5) = 152.80(1.6105)$$

$$= \$246.08$$

Practice Question

Tru-Test is in the business of assembling and packaging automotive and marine testing equipment to be sold through retailers to "do-it-yourselfers" and small repair shops. One of its products is tire pressure gauges. This operation has some excess capacity. Tru-Test is considering using this excess capacity to add engine compression gauges to its line. It can sell engine pressure gauges to retailers for $8 per gauge and expect to be able to produce about 1000 gauges in the first month of production. It also expects that, as the workers learn how to do the work more efficiently, productivity will rise by 0.25 percent per month for the first two years. In other words, each month's output of gauges will be 0.25 percent more than the previous month's. The interest rate is 1.5 percent per month. All gauges are sold in the month in which they are produced, and receipts from sales are at the end of each month. What is the present worth of the sales of the engine pressure gauges in the first two years?